

Lessons Learned Essay

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This project and course have deepened my understanding of UI design, evaluation, and implementation. Through the project, I gained hands-on experience addressing real-world application usability challenges and implementing solutions with high fidelity that meet user needs. Along the way, I learned how to gain valuable feedback from user research and user survey. I learned how to develop solutions to address users' frustration based on the feedback. I focused on enhancing existing functionality to increase usability while also adding new features to improve a better user experience. Our low fidelity prototype presented our design to solve the challenges and issues that users currently faced. While developing the design, I kept design principles and patterns in my mind. In term of evaluating our design, I learned to use cognitive walkthroughs and Heuristics evaluation. learned to refine my design based on findings from cognitive walkthroughs and heuristic evaluation. I applied these findings into the final high-fidelity prototype. During the user testing, I learned to identify the confusion points that user encountered. After the user testing, I evolved my design even better by simplifying the navigation sidebar and labeling, combining the driver and rider profile together, and improving user feedback to keep user notify every step along the way to complete their user tasks.

One of the most valuable lessons I learned was personas, user tasks and walkthrough scenarios. User tasks tells users what to do instead of how to do, which walkthrough scenarios tells user how to do step by step. This knowledge I can apply them to my other courses like Architecture Design and Project Management. When I design a new system architecture, I need to keep these in mind to help me to choose a propose solution can address user needs. I learned to empathize users' difficulty, and this is most important part in user research.

Another key takeaway was the significance of iterative feedback. Through heuristic evaluation and cognitive walkthroughs, I discovered how even small design decisions significantly impact usability. For example, unclear labels and missing error messages issues that were initially seemed minor but proved critical to the user experience. These evaluations highlighted the importance of rigorous interface testing and continuous design refinement based on user feedback. By simplifying the sidebar structure and maintaining consistent functionality across screens, I improved the application's usability while reinforcing the importance of prioritizing core user tasks. I also learned that implementing thoughtful feedback mechanisms, such as confirmations for actions and error messages, helps users feel confident and in control while interacting with the interface.

The project highlighted important trade-offs between innovation and feasibility. While we were excited to introduce ambitious features like carpooling contracts, electric payment systems, and driver license verification, their complexity led us to focus on implementing core functionalities.

Regarding the course format, I prefer the hybrid/flipped classroom model, as it allowed for flexible learning. If in the future the course can provide recorded lectures and live sessions together would be nice.

Overall, this course provided a comprehensive introduction to UI design and evaluation. I learned a tremendous amount from both the coursework and project. I'm proud that I originated the project's concept and contributed key features including the new login system, electronic payment integration, carpooling contracts, and driver license verification. I also developed the foundational user interface design concepts.

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